

Computer code TEMPA-SC: Computational results of the Benchmark Exercise on Flow Stability in Heated Channels with Supercritical Fluids

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Churkin – TEMPA-SC

Model Description

TEMPA-SC

Computer code TEMPA-SC [1,2] has been developed in OKB "GIDROPRESS" for steady-state and transient thermal-hydraulic analyses of supercritical water-cooled FA. The code is based on the TEMPA-1F code [3].

TEMPA-SC mathematical model is based on the 3D conservation equations written in an integral form for the sets of control volumes fixed in the space. These equations are: mass conservation, energy conservation and momentum conservation.

Discretization is typical for the sub-channel analysis. Cross section of the FA is divided into sets of parallel channels. The set of main channels (Figure 1a) is used for the mass, energy and longitudinal momentum conservation equations. The set of link channels (Figure 1b) is used for a transverse momentum conservation equation. A link channel connects two main channels. For example, the link channel ij connects main channel i and main channel j (Figure 1).

In the longitudinal direction the channels are divided into layers (Figure 2). The set of main layers is used for mass, energy and transverse momentum conservation equations. The set of link layers, placed between main layers, is used for a longitudinal momentum equation.

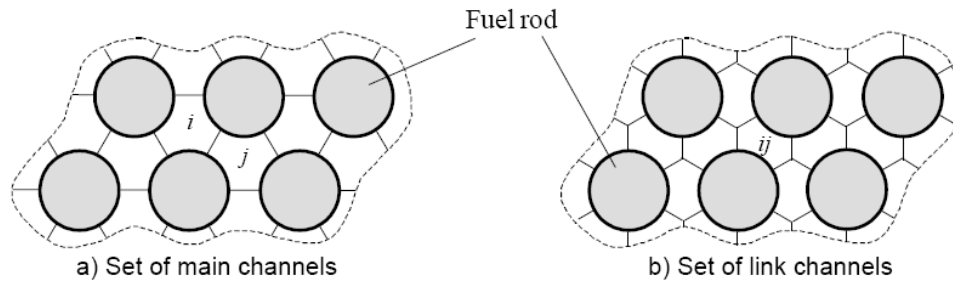


Figure 1. Fragmentation of FA cross-section

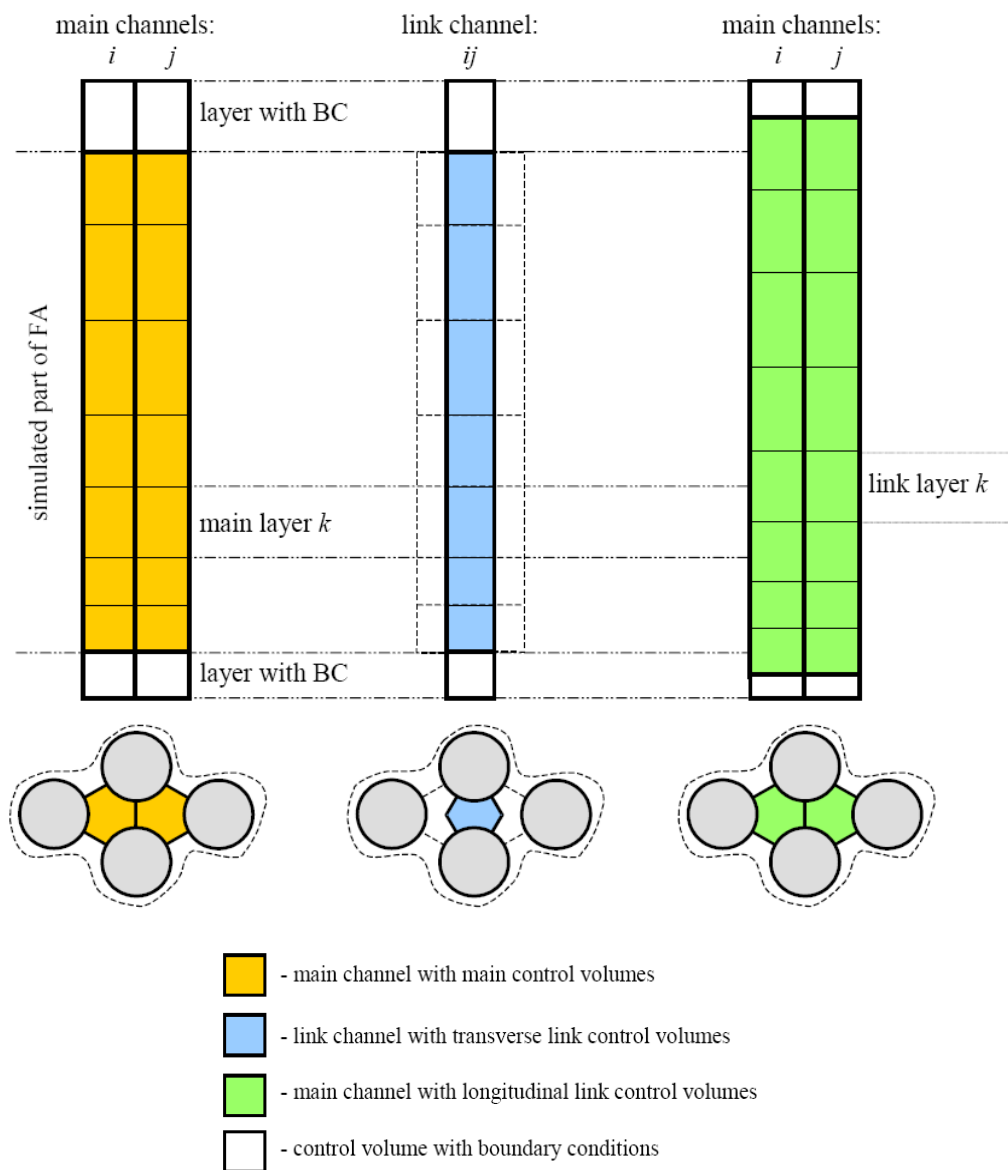


Figure 2. Fragmentation of the FA in longitudinal direction

Crossings of the channels and layers form three sets of control volumes:

$\{V_n\}$ - set of control volumes for mass and energy conservation equations;

$\{\tilde{V}_n\}$ - set of control volumes for longitudinal momentum conservation equation;

$\{\hat{V}_n\}$ - set of control volumes for transverse momentum conservation equations.

For these sets of the control volumes the base conservation equations are written as:

mass conservation equation:

$$\frac{d}{dt} \int_{V_n} \rho dV + \oint_{S_n} \rho v_i dS_i = M_{src_n}, \quad (1)$$

energy conservation equation:

$$\frac{d}{dt} \int_{V_n} \rho E dV + \oint_{S_n} \rho v_i E dS_i = \int_{V_n} \rho v_i f_i dV - \oint_{S_n} q_i dS_i - \oint_{S_n} v_i p dS_i + \oint_{S_n} \sigma_{ij} v_i dS_j + (E_{src} M_{src})_n, \quad (2)$$

longitudinal momentum conservation equation:

$$\frac{d}{dt} \int_{\tilde{V}_n} \rho v_z dV + \oint_{\tilde{S}_n} \rho v_z v_j dS_j = \int_{\tilde{V}_n} \rho f_z dV - \oint_{\tilde{S}_n} p dS_z + \oint_{\tilde{S}_n} \sigma_{zj} dS_j, \quad (3)$$

transverse momentum conservation equations:

$$\frac{d}{dt} \int_{\hat{V}_n} \rho v_i dV + \oint_{\hat{S}_n} \rho v_i v_j dS_j = \int_{\hat{V}_n} \rho f_i dV - \oint_{\hat{S}_n} p dS_i + \oint_{\hat{S}_n} \sigma_{ij} dS_j, \quad (i = x, y). \quad (4)$$

where S_n – is surface of the control volume V_n ; $\tilde{S}_n$ – is surface of the control volume $\tilde{V}_n$; $\hat{S}_n$ – is surface of the control volume $\hat{V}_n$.

After some conversions the system of equations (1) – (4) is transformed into:

$$\frac{d}{dt} (\rho_{ik} V_{ik}) + \tilde{G}_{ik} - \tilde{G}_{ik-1} + \sum_j M_{ijk} + \Theta_{ik}^p = M_{src_{ik}}, \quad (5)$$

$$\frac{d}{dt} (\rho_{ik} h_{ik} V_{ik}) + \tilde{G}_{ik} \tilde{h}_{ik} - \tilde{G}_{ik-1} \tilde{h}_{ik-1} + \sum_j M_{ijk} h_{ijk} + \Theta_{ik}^h = \sum_w q_{ik}^w S_{ik}^w + (h_{src} M_{src})_{ik}, \quad (6)$$

$$\begin{aligned} \frac{d}{dt} (\tilde{G}_{ik} \tilde{l}_k) + G_{ik+1} w_{ik+1} - G_{ik} w_{ik} + \sum_j \tilde{M}_{ijk} \tilde{w}_{ijk} + \tilde{\Theta}_{ik}^G = \tilde{\rho}_{ik} f_z \tilde{V}_{ik} + \\ + p_{ik} F_{ik} - p_{ik+1} F_{ik+1} - \tilde{p}_{ik} (F_{ik} - F_{ik+1}) - \tilde{\tau}_{ik} \tilde{S}_{ik}^w, \end{aligned} \quad (7)$$

$$\begin{aligned} \frac{d}{dt} (M_{ijk} l_{ij}) + \tilde{G}_{ijk} \tilde{v}_{ijk} - \tilde{G}_{ijk-1} \tilde{v}_{ijk-1} + (\rho_{jk} (\tilde{v}_{jk} \tilde{n}_{ij})^2 - \rho_{ik} (\tilde{v}_{ik} \tilde{n}_{ij})^2) S_{ijk} = \\ = \rho_{ijk} \tilde{f} \tilde{n}_{ij} V_{ijk} + (p_{ik} - p_{jk}) S_{ijk} - \tau_{ijk} S_{ijk}^w. \end{aligned} \quad (8)$$

Basic decision variables ρ_{ik} , E_{ik} , $\tilde{G}_{ik}$ и M_{ijk} are average for the control volumes.

The following assumptions are used to get equations (5) – (8):

- all decision variables refer to the centres of control volumes;
- fluid velocity at the wall is equal to zero;
- in the centre of the transverse link control volume the fluid velocity projection onto the line between the centres of rods is equal to zero;
- longitudinal diffusion is neglected;
- transverse diffusion of the transverse momentum in the equation (8) is neglected;
- the conversion of internal energy E into enthalpy h .

To solve the system of equations (5) – (8) boundary conditions should be given for each main channels. Inlet boundary conditions are:

- pressure or flowrate;
- enthalpy or temperature.

Outlet boundary conditions are:

- pressure;
- enthalpy or temperature.

The user can select different correlations for the calculation of the friction factor and heat transfer coefficients.

Two thermophysical properties of water ρ and h are solutions of the equations (5) – (8). Two different methods can be used to calculate p and T . First method uses table of properties $\rho=\rho(p,T)$ and $h=h(p,T)$. This method applies special algorithm for finding p and T on the known values of ρ and h . The ranges of key parameters are: $23 \text{ MPa} < p < 49 \text{ MPa}$; $0 < T < 800^\circ\text{C}$. This method has high accuracy, but significantly increases the calculation time.

Second method is based on approximation of the thermophysical properties of water. This method is faster than the first method, but its ranges of key parameters are narrower: $23.5 \text{ MPa} < p < 26 \text{ MPa}$; $20^\circ\text{C} < T < 800^\circ\text{C}$. Also second method is less accurate than the first method.

For the numerical solution of differential equations (5) – (8) used technique based on the FLIC method [4].

Additional information about the code TEMPA-SC is in [1-3].

Methodology of the analysis

The transient calculations were used to define the stability threshold. Of the cases presented in the [5] were considered only the cases with water.

In the calculation of one case, the fluid boundary conditions (pressure and temperature) were constant. Heat flux to the coolant was assigned as a function of time. The boundary conditions for the considered cases are provided in Table 1.

In accordance with [6] for the cases at $K_{out}=20$ the pressure drop is assigned to be equal to 0.12 MPa. At this, for the subcase with $N_{SPC}=1.5$ coolant flowrate through the channel when it is not heated, is equal to 0.091 kg/s. For cases with other values of K_{out} the pressure drop was chosen proceeding from the condition that the flowrate is equal to 0.091 kg/s at $N_{SPC}=1.5$.

Table 1. Boundary conditions for the calculations cases

Cases		Boundary conditions	
K_{out}	N_{SPC}	T_{in} , °C	$\Delta p_{ch}^{*})$, Pa
20	0.5	375	$0.12 \cdot 10^6$
	1.0	340	
	1.5	286	
	2.0	223	
	2.5	154	
10	1.5	286	$0.10 \cdot 10^6$
5	1.5	286	$0.93 \cdot 10^5$
2	0.5	375	$0.87 \cdot 10^5$
	1.0	340	
	1.5	286	
	2.0	223	
	2.5	154	
* $p_{out} = 25 \text{ MPa}$; $p_{in} = p_{out} + \Delta p_{ch}$			

In the course of the transient as the fluid temperature increased at the channel outlet, the flowrate through the channels decreased, which caused a still greater temperature increase. At a certain moment the flow became unstable and oscillations of the coolant flowrate arose. Close to the instability threshold the rate of power increasing was 21,3 W/s, but at the

beginning of the transient within 10 s (from the 2nd to the 12th seconds) a considerable power surge was assigned which reduced the calculation time. The plot of the power increasing is schematically presented in Figure 3.

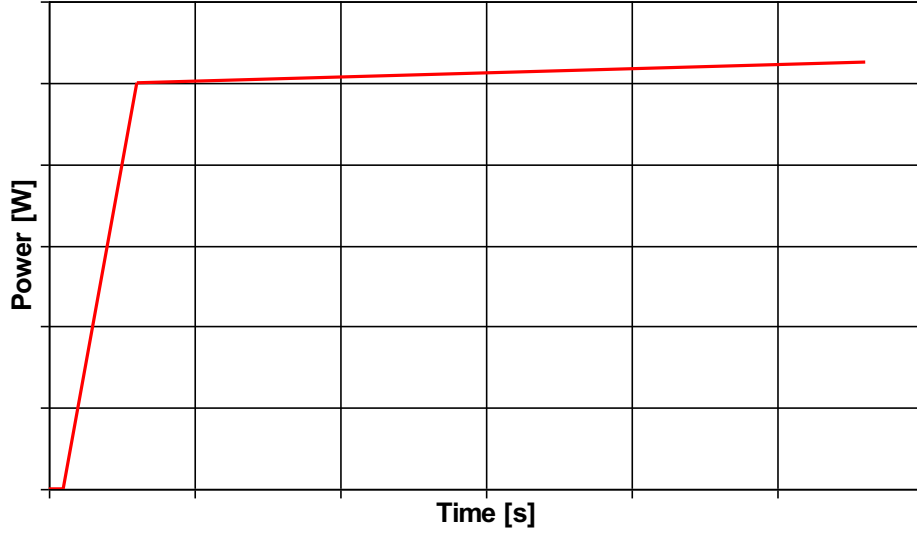


Figure 3. Scheme of the power increasing

The pipe was modeled as one channel, divided into 61 nodes.

Close to the instability threshold next formal criteria was used to define the beginning of instability:

$$\frac{dN_{TPC}}{dt} = 0.$$

Obtained results for the base cases with water

The plots of heating power, inlet and outlet flow rate, N_{TPC} as a function of time for the different cases are presented in Figures 4 – 11.

For cases with $K_{out}=2$ all calculations were automatically interrupted before the instability threshold because outlet water temperature amount to 800°C (code limitation) (see Figure 12).

All calculation results are summarized in Stability Map (see Table 2 and Figure 13).

Table 2. Results of calculations: instability threshold

Cases		Results
K_{out}	N_{SPC}	N_{TPC}
20	0.5	3.21
	1.0	2.8
	1.5	2.93
	2.0	3.35
	2.5	3.73
10	1.5	3.53
5	1.5	4.28
2	0.5	-
	1.0	-
	1.5	-
	2.0	-
	2.5	-

The plots of the period of oscillations and ratio of transit time to the period of oscillations are presented in Figures 14, 15.

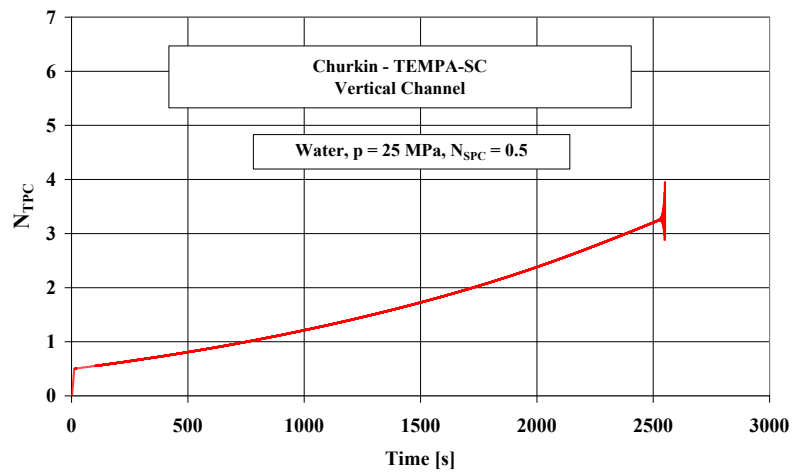
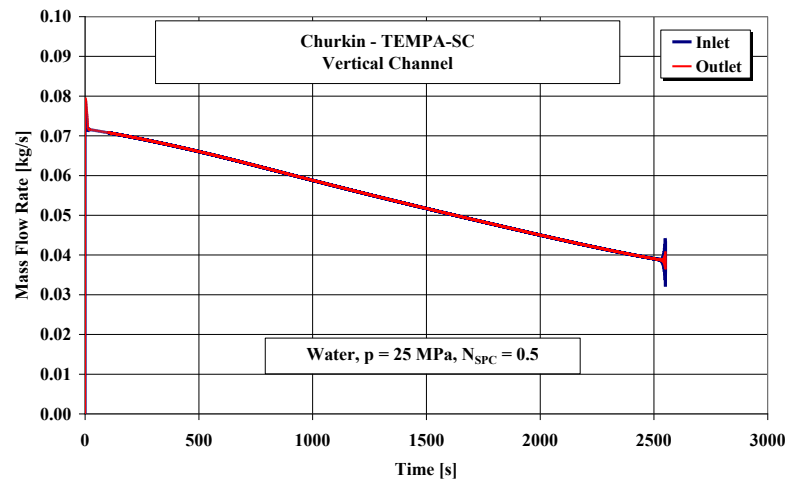
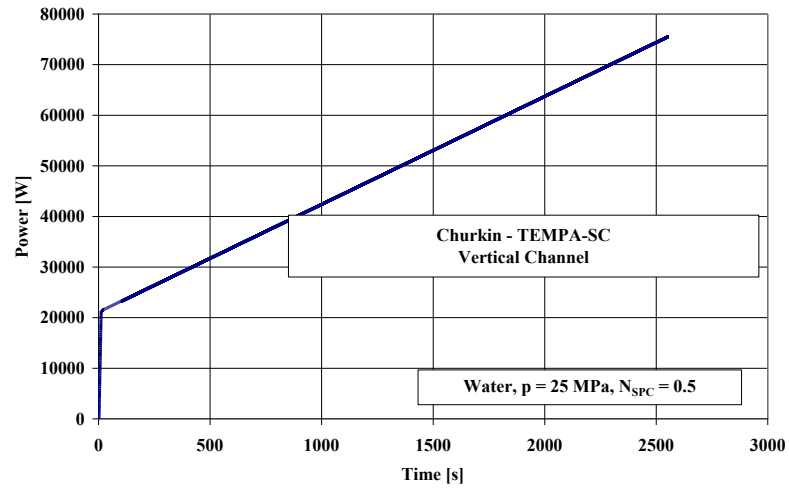


Figure 4. Calculation results for the case with $K_{out}=20$ and $N_{spc}=0.5$

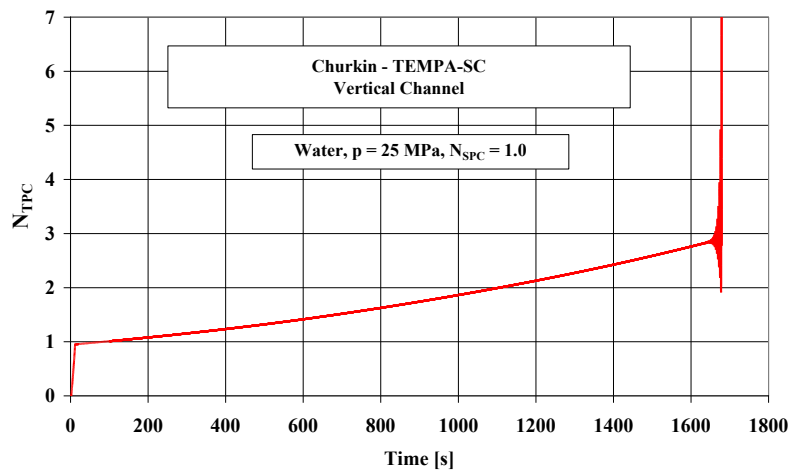
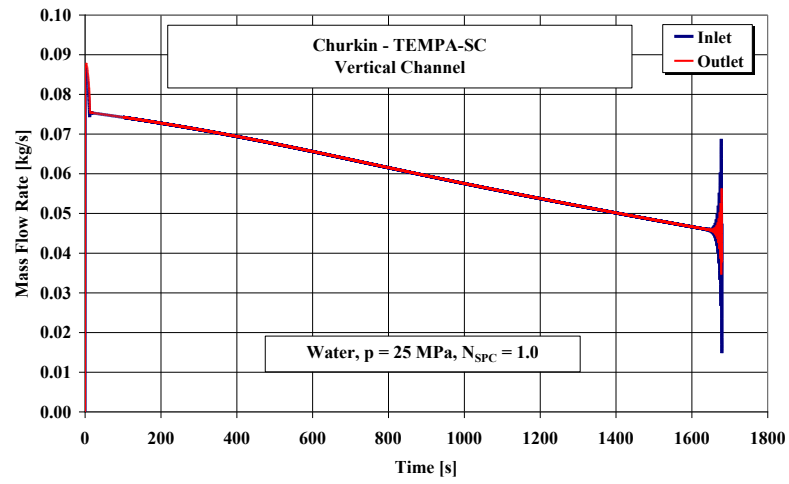
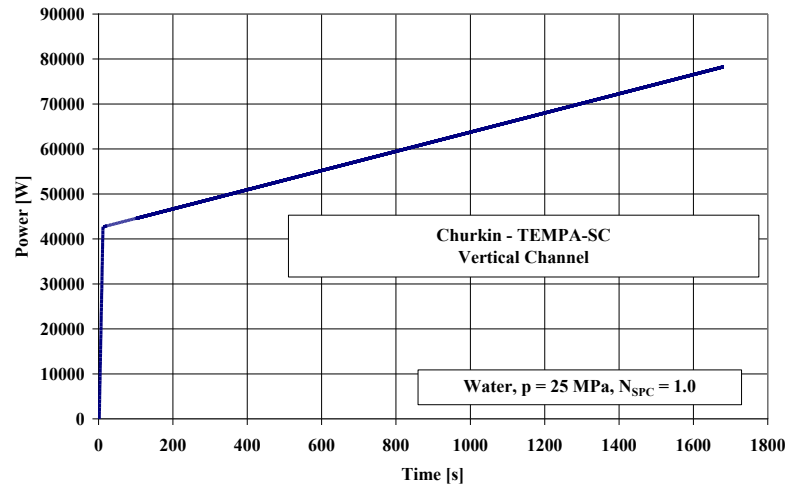


Figure 5. Calculation results for the case with $K_{out}=20$ and $N_{spc}=1.0$

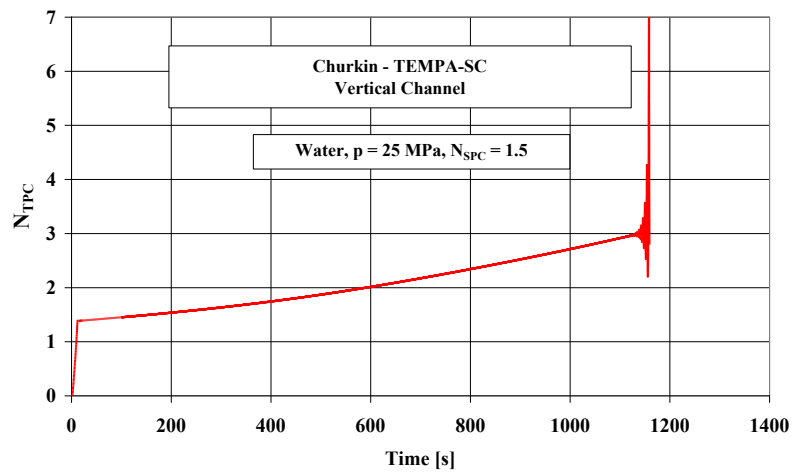
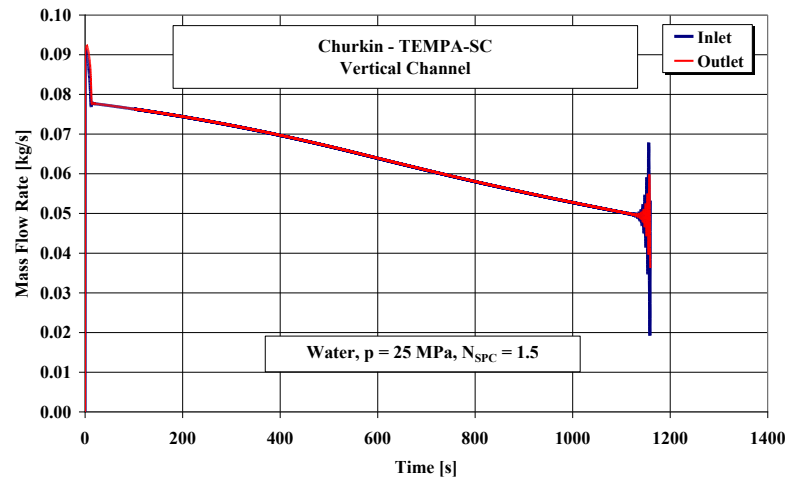
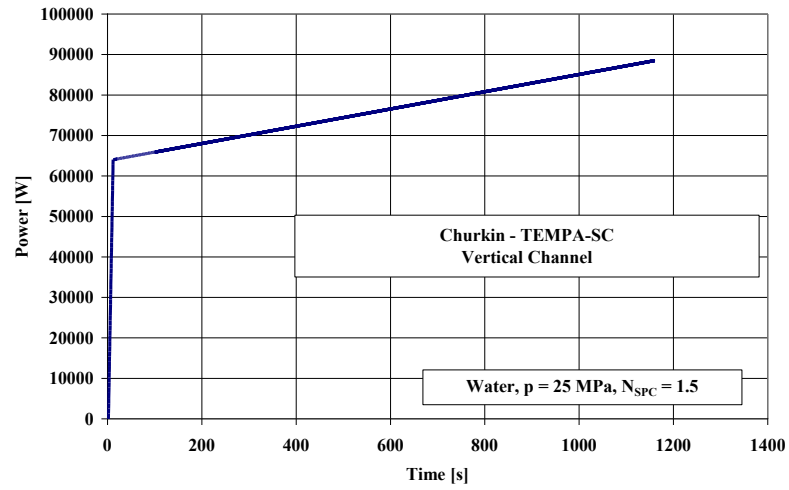


Figure 6. Calculation results for the case with $K_{\text{out}}=20$ and $N_{\text{spc}}=1.5$

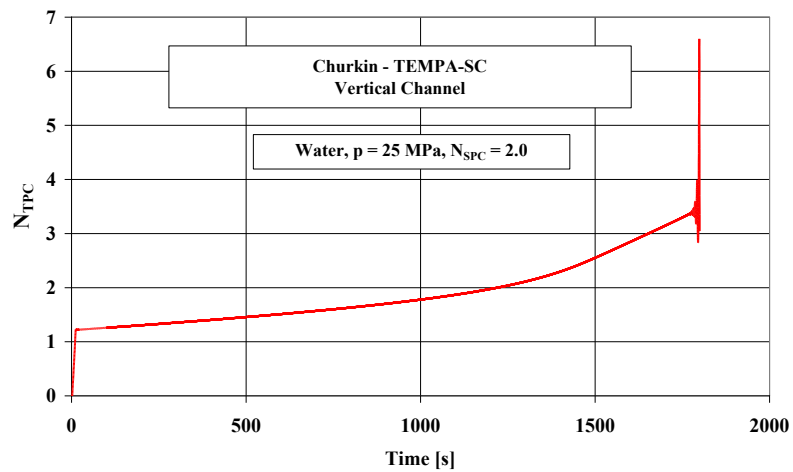
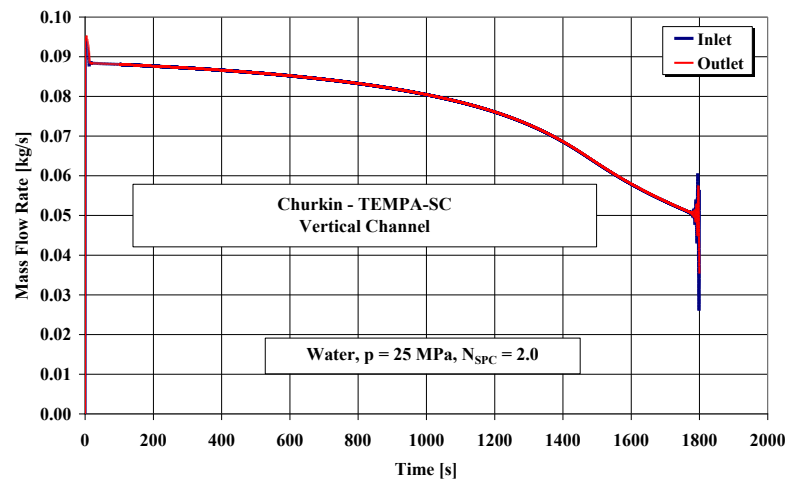
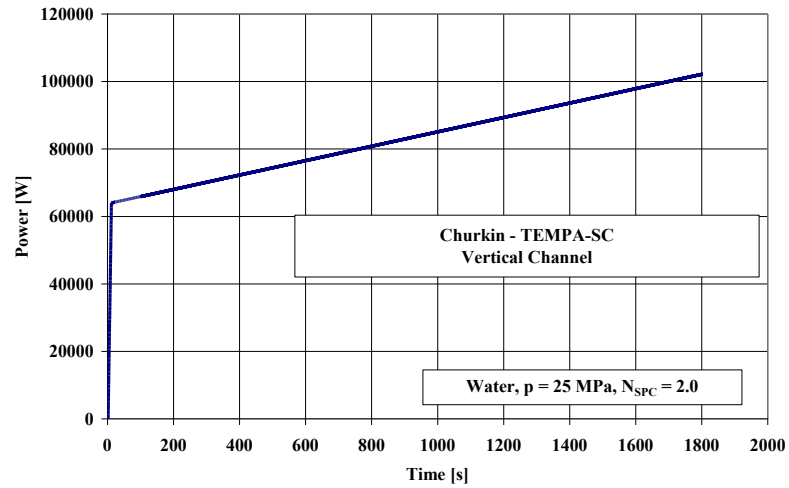


Figure 7. Calculation results for the case with $K_{\text{out}}=20$ and $N_{\text{spc}}=2.0$

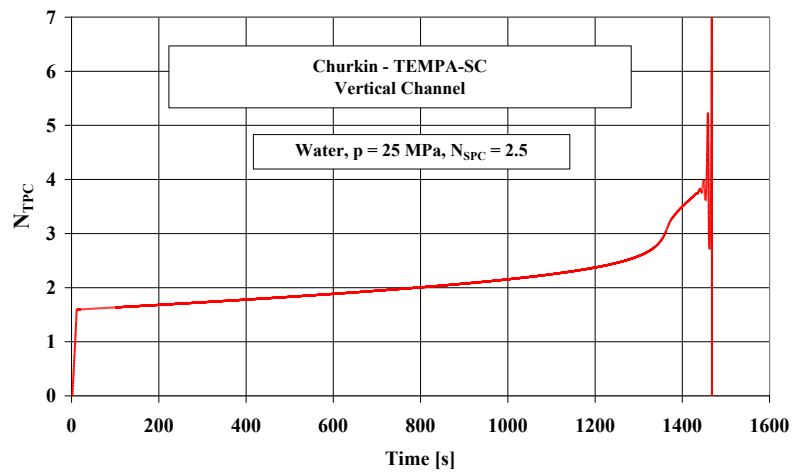
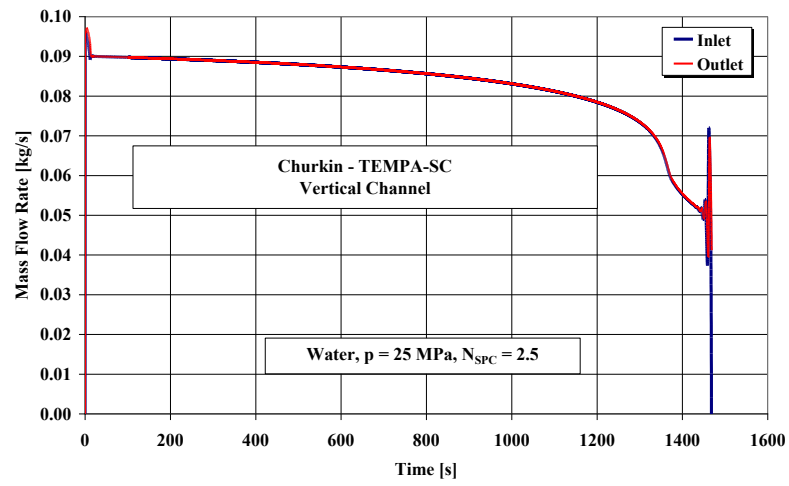
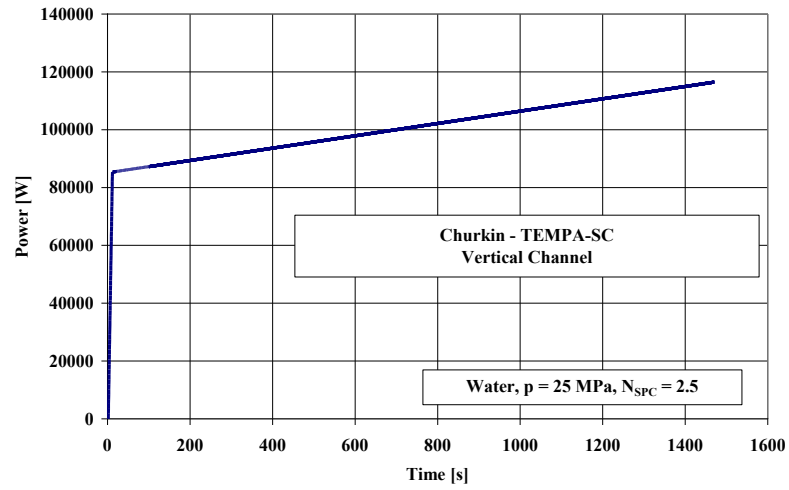


Figure 8. Calculation results for the case with $K_{\text{out}}=20$ and $N_{\text{spc}}=2.5$

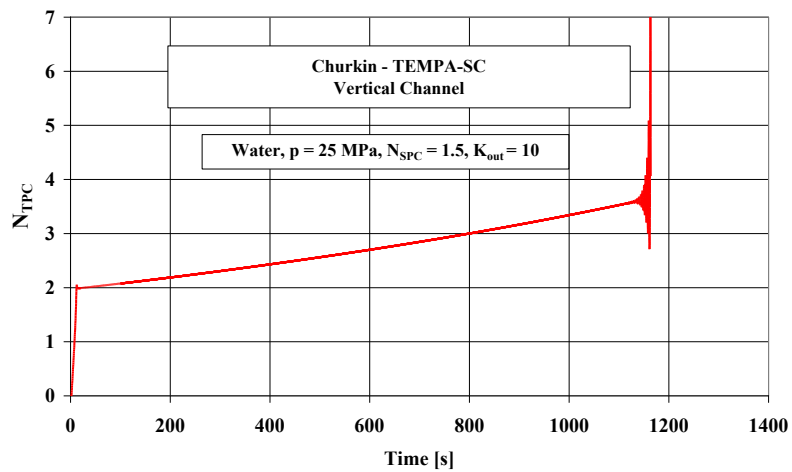
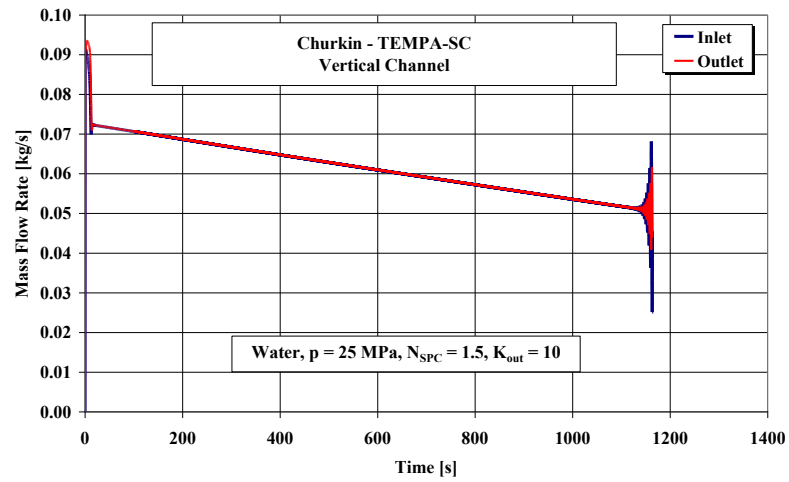
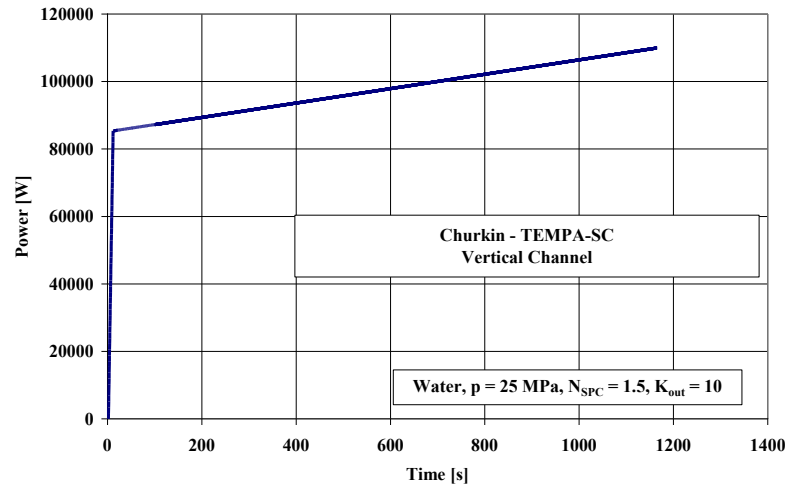


Figure 9. Calculation results for the case with $K_{out}=10$ and $N_{spc}=1.5$

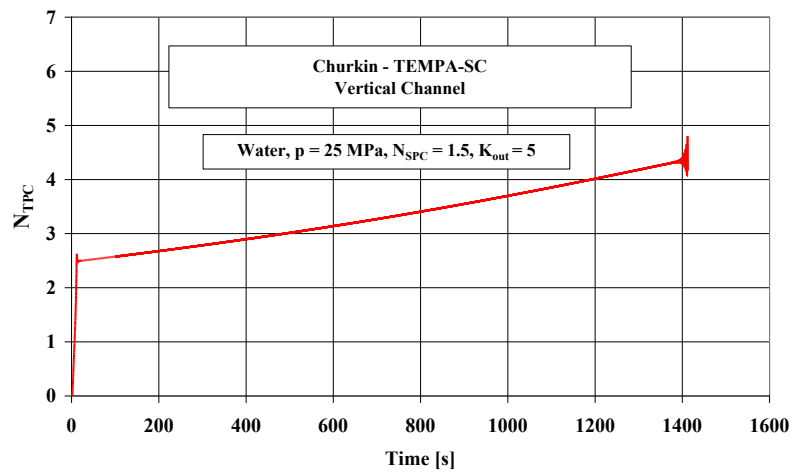
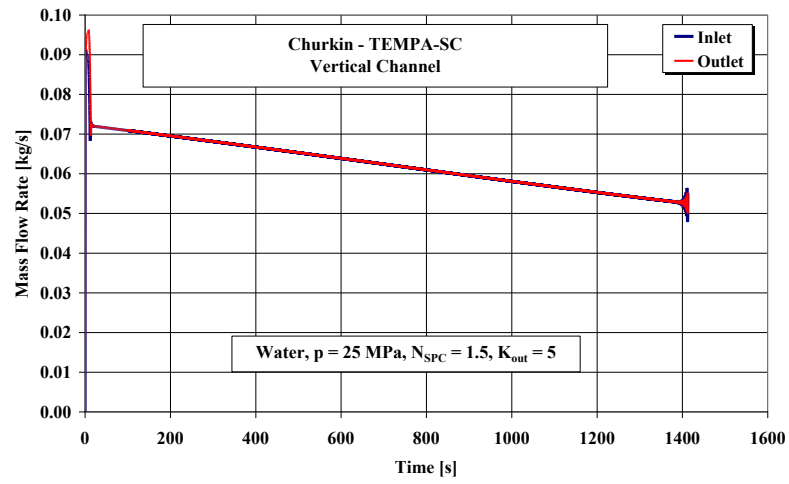
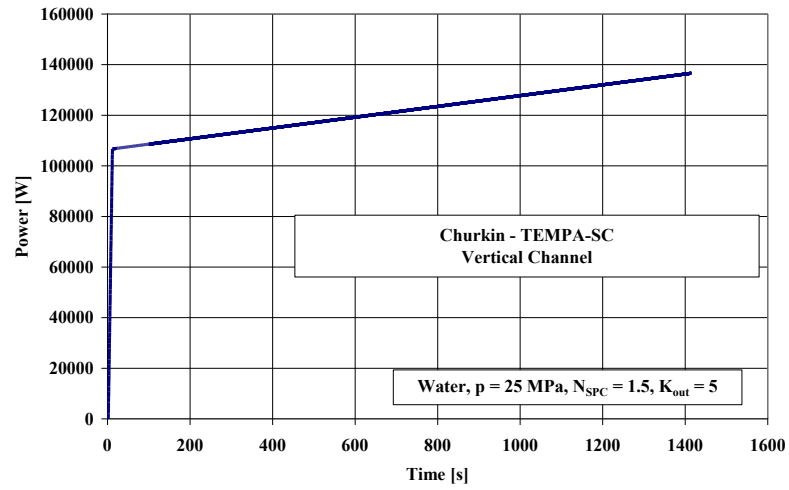


Figure 10. Calculation results for the case with $K_{out}=5$ and $N_{spc}=1.5$

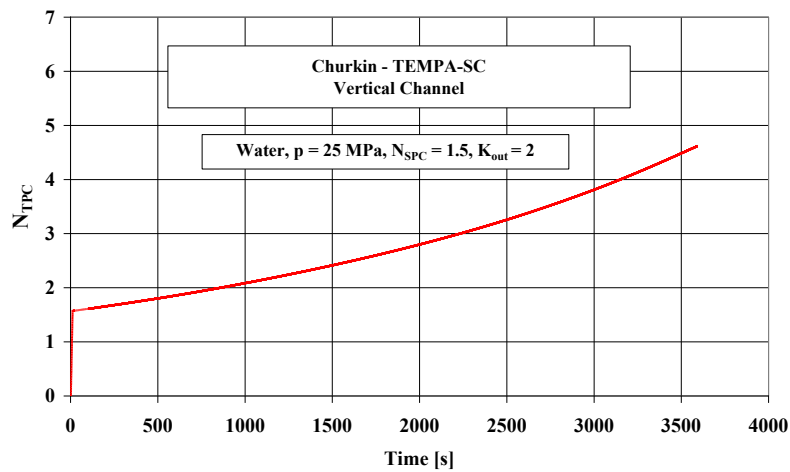
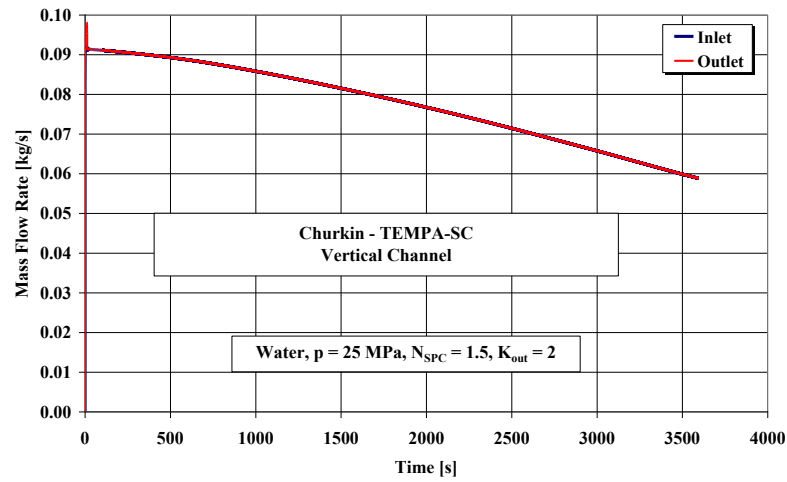
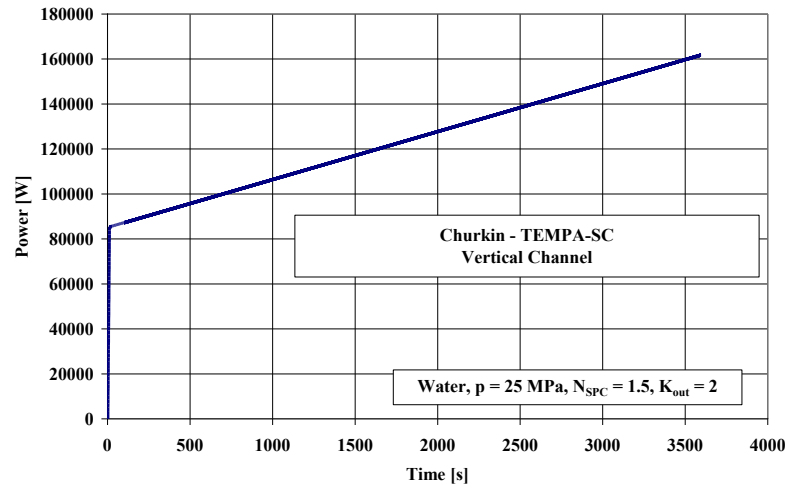


Figure 11. Calculation results for the case with $K_{out}=2$ and $N_{spc}=1.5$

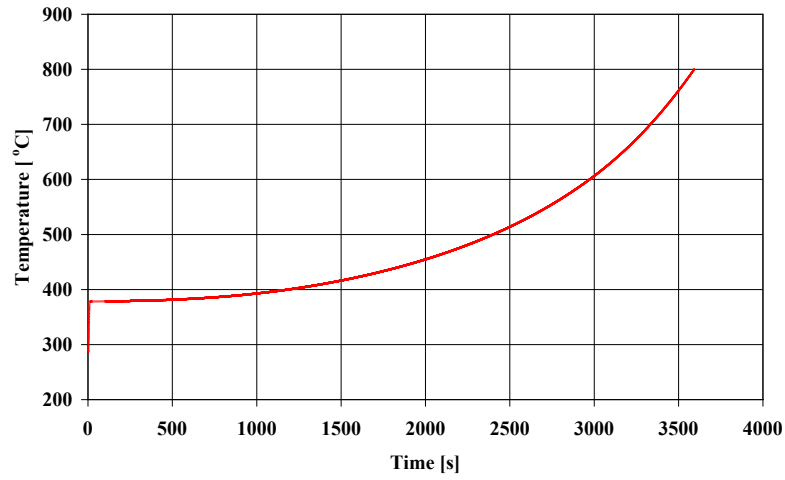


Figure 12. Output fluid temperature for the case with $K_{out}=2$ and $N_{spc}=1.5$

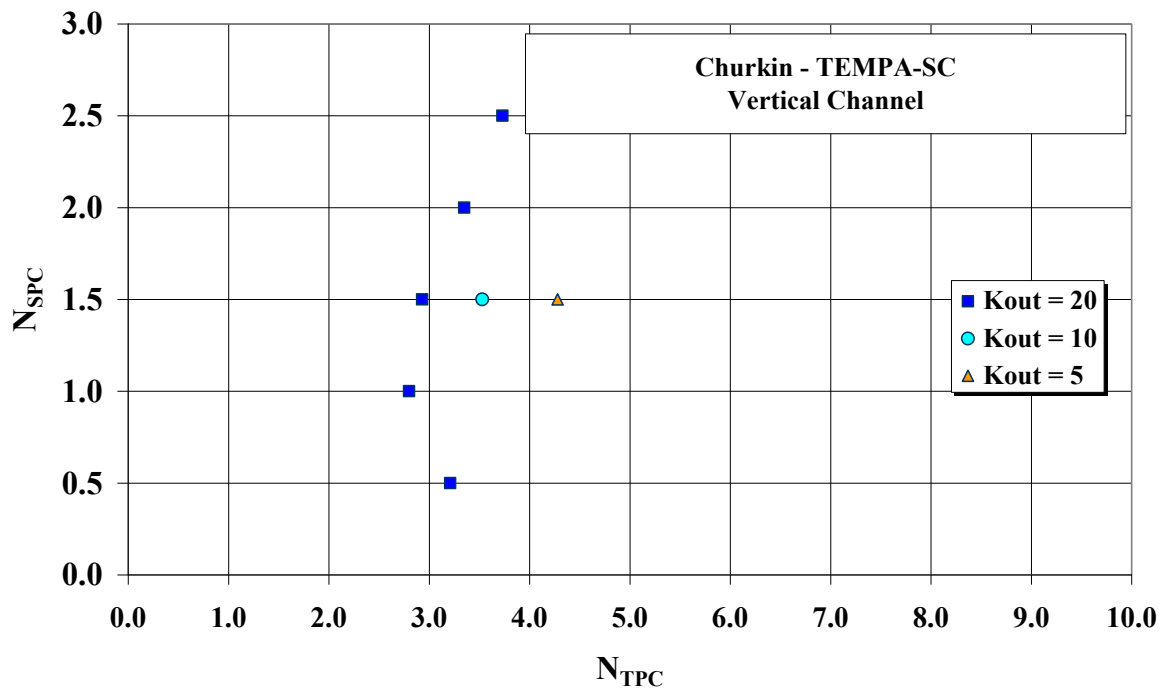


Figure 13. Stability Map for the vertical channel with water

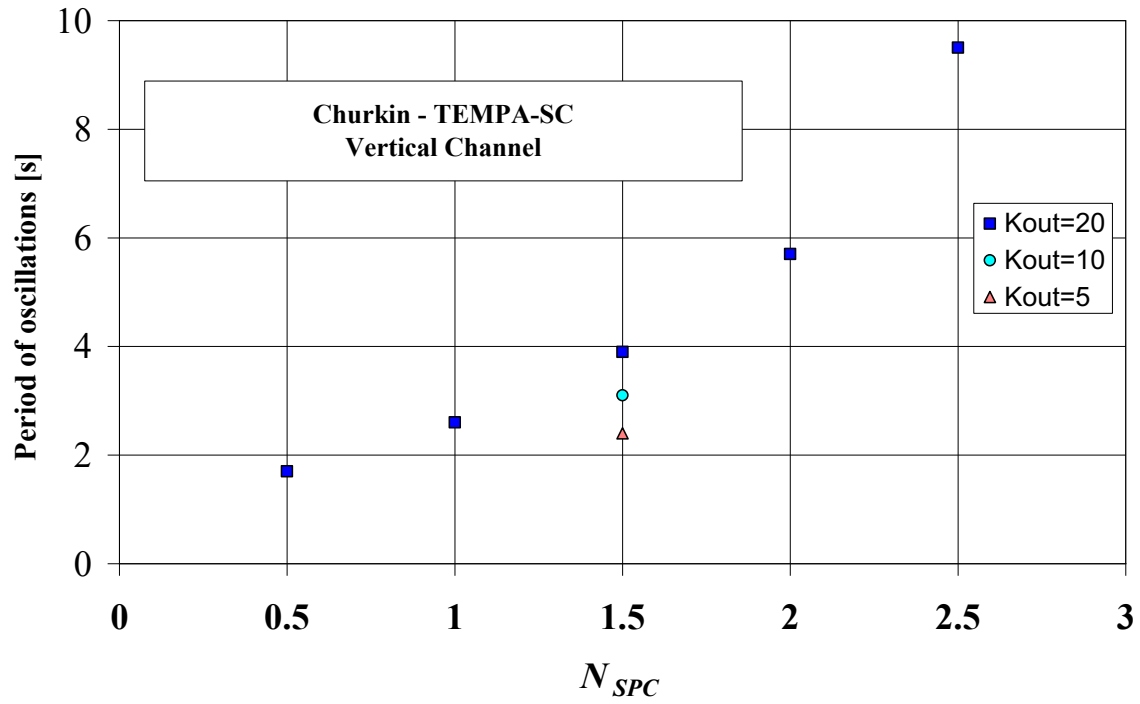


Figure 14. Period of oscillations as function of N_{SPC}

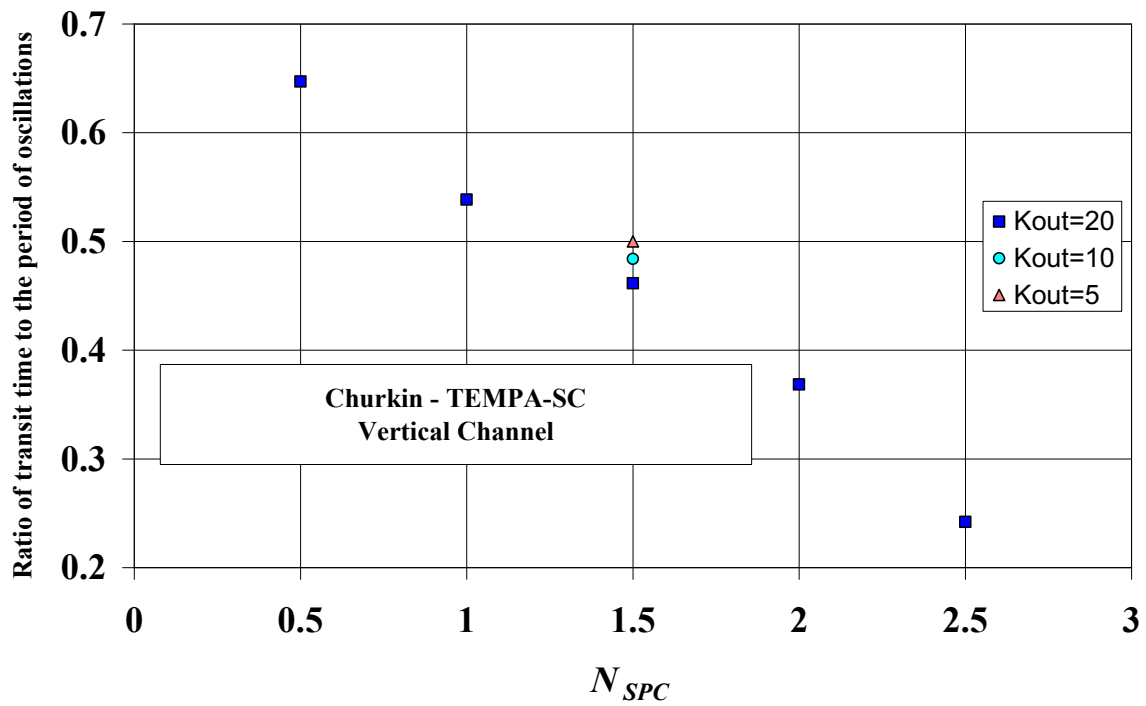


Figure 15. Ratio of transit time to the period of oscillations as function of N_{SPC}

Sensitive analysis and inaccuracy estimation

It will be submitted at the 3rd RCM.

References

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Nomenclature

E – internal energy per unit mass, J/kg;

F – cross section area, m²;

f – body force per unit mass, m/s²;

G – longitudinal coolant flowrate, kg/s;

g – acceleration of gravity, m/s²;

h – enthalpy, J/kg;

l – length, m;

M – transfer coolant flowrate, kg/s;

N_{SPC} – sub-pseudocritical number;
 N_{TPC} – trans-pseudocritical number;
 n – outward unit vector to the surface S ;
 p – pressure, Pa;
 q – heat flux, W/m²;
 S – surface area, m²;
 T – temperature, °C;
 t – time, s;
 V – volume, m³;
 v – transverse velocity, m/s;
 w – longitudinal velocity, m/s;
 z – axial distance, m.

Greek letters

Θ^G – turbulent flux of the longitudinal momentum, kg·m/s²;
 Θ^h – turbulent flux of the enthalpy, W;
 Θ^p – turbulent flux of the mass, kg/s;
 ρ – density, kg/m³;
 τ – wall shear stress, N/m²;
 σ – viscous stress tensor, N/m².

Subscripts

ch – channel;
 i, j – channel number; summation index of coordinate in equations (1) – (4);
 k – layer number;
 n – control volume number;
 w – wall;
 z – in axial direction;
 in – inlet;
 out – outlet;
 src – coolant source.